

LIANG YU

www.linkedin.com/in/liangyu419

[www.github.com/yuliang419](https://github.com/yuliang419)

German and South African dual citizen



EDUCATION

Doctor of Philosophy | *Physics*

Massachusetts Institute of Technology (MIT)

Jun 2019

Cambridge, MA, USA

- Thesis: Elucidating the Evolution Histories of Giant Planets with *K2* and *TESS*
- Research areas: Exoplanet detection and characterization
- GPA: 5.00 / 5.00

Bachelor of Science | *Astronomy & Physics*

Yale University

May 2014

New Haven, CT, USA

- Graduated summa cum laude, with exceptional distinction in the Astronomy & Physics major
- GPA: 4.00 / 4.00

RELEVANT EXPERIENCE

Lead AI Architect

BMW Group

Nov 2024 – present

Munich, Germany

- Lead the development of onboard machine learning platform to support custom inference frameworks, scheduling, agentic workflows and monitoring on constrained in-vehicle hardware.
- Evaluate and benchmark emerging generative AI models and accelerators, with a focus on deploying small language models (SLMs) on NPUs for production-grade automotive use cases.
- Fine tune and optimize SLMs for heterogeneous hardware and frameworks for in-vehicle voice assistant use cases. Ensure a balance between efficient execution and reliability.
- Lead hiring and organizational development for the ML Core Platform team in India, including technical mentorship and performance management.
- Advise executive leadership and board stakeholders on AI strategy, influencing investment decisions and product direction across global markets, with deep focus on China (technology trends, competitors, and regulatory landscape).

Data Science Team Lead, Big Loop and Advanced Systems

Volkswagen Group CARIAD SE (Audi AG until Apr 2021)

Oct 2020 – Oct 2024

Ingolstadt, Germany

- Hired and supervised a group of 15+ software engineers and data scientists. Established engineering best practices, development workflows, and a production-grade codebase for data-centric AI systems for autonomous driving.
- Led the development of in-vehicle data selection triggers for camera images, enabling efficient acquisition of high-value training data by detecting high-uncertainty and novel objects in real-world driving scenarios.
- Defined product vision and roadmap for data selection triggers, communicated with cross-functional stakeholders and oversaw successful deployment of triggers across the VW Group fleet.
- Contributed as data scientist and developer to the research, testing and optimization of uncertainty quantification and anomaly detection algorithms to solve the “long-tail” problem, as well as active learning experiments for perception tasks (semantic segmentation, object detection).
- Influenced executive decision-making by developing statistical models to estimate the required empirical test effort for autonomous vehicles.

Data Scientist, Upstream Integrated Solutions

ExxonMobil

Aug 2019 – Sep 2020

Spring, TX, USA

- Developed smoothing filter-based models for the calibration and interpretation of time series data from acoustic sand monitors and seismic sensors, and a web API for visualizing sensor data and sending real-time alerts to users.
- Automated predictions of drilling outcomes of basins using supervised learning. Matched the accuracy of predictions by experts in geophysics.
- Researched and applied polarization filtering and spectral analysis to detect low signal-to-noise microseismic events in time series data from three-component seismic sensors.

Graduate Research Student, Exoplanet Group

MIT, Department of Physics

Sep 2014 – Jun 2019

Cambridge, MA, USA

- Co-authored [Astronet-Triage](#) and [Astronet-Vetting](#), convolutional neural networks that identify planet candidates and false positives in time series data from large-scale surveys. Currently used by the Transiting Exoplanets Survey Satellite (TESS) mission.
- Constructed pipeline to extract stellar flux, remove systematic noise with Gaussian process regression and identify planet-like signals from raw data taken by the K2 space mission. Discovered 81 confirmed planets and candidates.

Intern, Big Data Strategies

Totsa Total Oil Trading SA

Jun 2016 – Aug 2016

Geneva, Switzerland

- Researched supervised learning techniques to improve ship tracking accuracy, predict ship destinations based on partial trajectories, and identify patterns and anomalies in global ship movement.
- Developed algorithms to extract the most common ship trajectories between ports and calculate the length and duration of a typical trip.
- Simplified ship trajectories and removed outliers with clustering methods for easier processing.

STUDENT SUPERVISION**ElHayani, M. Master's thesis — Technical University of Munich**

2023-2024

Title: Real-Time Object Detection Uncertainty Quantification Using Augmented Images for Autonomous Vehicles

TECHNICAL SKILLS

Data science/ML: Edge AI, computer vision, generative AI (LLM/VLM), agentic AI, RAG, statistics, Bayesian inference, uncertainty quantification, time series analysis

Programming: Experienced with Python (incl. PyTorch, Keras, TensorFlow, Scikit-learn, LangChain, LangGraph) and SQL; basic knowledge of Bash, R, MATLAB and Mathematica

Technologies: Git, Azure

Languages: fluent English and Mandarin Chinese; intermediate German, French, Spanish, and Italian

INVITED TALKS

Finding the Unknown Unknowns: Intelligent Data Collection for Autonomous Driving Development

2022

WeAreDevelopers World Congress

An Uncertainty-Based Corner Case Detection Framework for Autonomous Driving

2021

Sensors & IIoT - Manufacturing, Automation and Robotics

Model-Enhanced Machine Learning for Astronomical Discovery

2016

University of Toronto

Tests of the Planetary Hypothesis for PTFO 8-8695b

2015

Boston University

Tests of the Planetary Hypothesis for PTFO 8-8695b

2015

Harvard University

SELECTED PUBLICATIONS

Yu, L.; Vanderburg, A.; Huang, C. X.; Shallue, C. J.; Crossfield, I. J. M. et al. (2019), *Identifying Exoplanets with Deep Learning III: Automated Triage and Vetting of TESS Candidates*, the Astronomical Journal, 158, 25.

Yu, L.; Zhou, G.; Rodriguez, J. E.; Huang, C. X.; Vanderburg, A. et al. (2018), *EPIC 246851721 b: A Tropical Jupiter Transiting a Rapidly Rotating Star in a Well-Aligned Orbit*, the Astronomical Journal, 156, 250.

Yu, L.; Rodriguez, J. E.; Eastman, J. D.; Crossfield, I. J. M.; Shporer, A. et al. (2018), *Two Warm, Low-density Sub-Jovian Planets Orbiting Bright Stars in K2 Campaigns 13 and 14*, the Astronomical Journal, 156, 127.

Yu, L.; Crossfield, I. J. M.; Schlieder, J. E.; Kosiarek, M. R.; Feinstein, A. D. et al. (2018), *Planetary Candidates from K2 Campaign 16*, the Astronomical Journal, 156, 22.

Yu, L.; Winn, J. N.; Gillon, M.; Albrecht, S.; Rappaport, S. et al. (2015), *Tests of the Planetary Hypothesis for PTFO 8-8695b*, the Astrophysical Journal, 812, 48.

Yu L.; Nelson, K.; Nagai, D. (2015), *The Influence of Mergers on Scatter and Evolution in Sunyaev-Zel'dovich Effect Scaling Relations*, the Astrophysical Journal, 807, 12.